HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

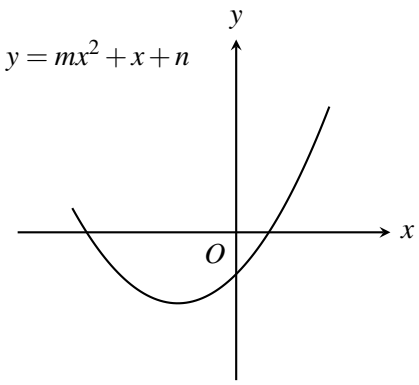
Student No.: 05 Name: _____ Marks: / 24

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The figure shows the graph of $y = mx^2 + x + n$, where m and n are constants. Which of the following is true?



A. $m < 0$ and $n < 0$

B. $m < 0$ and $n > 0$

C. $m > 0$ and $n < 0$

D. $m > 0$ and $n > 0$

(DSE-2014 PII Q05)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. If $a > b$ and $k < 0$, which of the following must be true?

I. $a^2 > b^2$

II. $a + k > b + k$

III. $\frac{a}{k^2} > \frac{b}{k^2}$

A. I only

B. II only

C. I and III only

D. II and III only

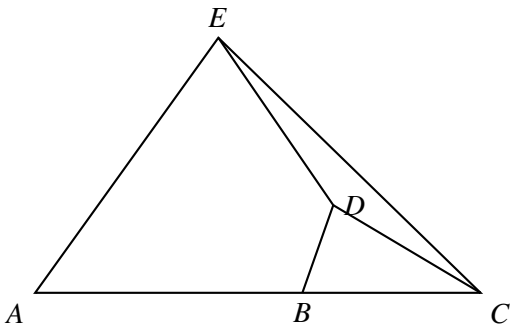
(DSE-2014 PII Q06)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. In the figure, B is a point lying on AC such that $AB : BC = 3 : 2$. It is given that $AE \parallel BD$. If the area of $\triangle BCD$ and the area of $\triangle CDE$ are 4 cm^2 and 8 cm^2 respectively, then the area of the trapezium $ABDE$ is



A. 18 cm^2.

B. 21 cm^2.

C. 27 cm^2.

D. 33 cm^2.

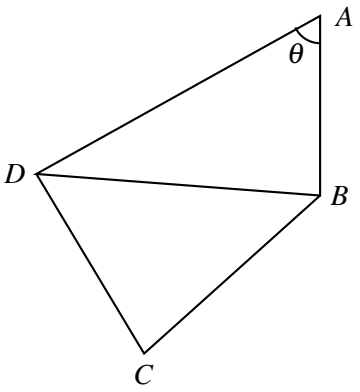
(DSE-2014 PII Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. In the figure, $\angle ABD = \angle ADC = \angle BCD = 90°$. If $AB = \ell$, then $CD =$



A. $\ell \sin\theta$.

B. $\ell \cos\theta$.

C. $\ell \sin\theta \tan\theta$.

D. $\dfrac{\ell \tan\theta}{\cos\theta}$.

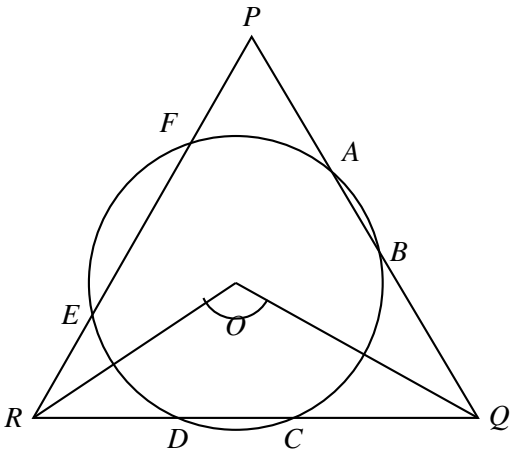
(DSE-2014 PII Q18)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. In the figure, O is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at A, B, C, D, E and F. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$



A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. The rectangular coordinates of the point P are $(-1, \sqrt{3})$. If P is reflected with respect to the x-axis, then the polar coordinates of its image are

A. $(2, 210°)$.

B. $(2, 240°)$.

C. $(4, 210°)$.

D. $(4, 240°)$.

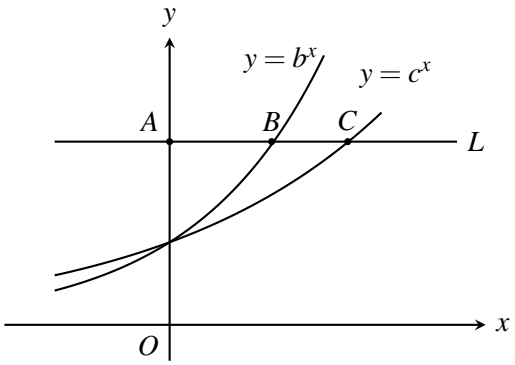
(DSE-2014 PII Q23)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. The figure shows the graph of $y = b^x$ and the graph of $y = c^x$ on the same rectangular coordinate system, where b and c are positive constants. If a horizontal line L cuts the y-axis, the graph of $y = b^x$ and the graph of $y = c^x$ at A, B and C respectively, which of the following are true?



I. $b < c$

II. $bc > 1$

III. $\dfrac{AB}{AC} = \log_b c$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q32)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. Which of the following is the greatest?

A. 124^{241}

B. 241^{214}

C. 412^{142}

D. 421^{124}

(DSE-2014 PII Q33)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. For $0° \leq x \leq 360°$, how many roots does the equation $7\sin^2 x = \sin x$ have?

A. 2

B. 3

C. 4

D. 5

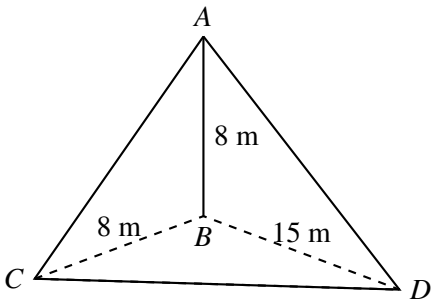
(DSE-2014 PII Q39)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. In the figure, AB is a vertical pole standing on the horizontal ground BCD, where $\angle CBD = 90°$. If the angle between the plane ACD and the horizontal ground is θ, then $\tan\theta =$



A. $\frac{8}{15}$.

B. $\frac{15}{8}$.

C. $\frac{15}{17}$.

D. $\frac{17}{15}$.

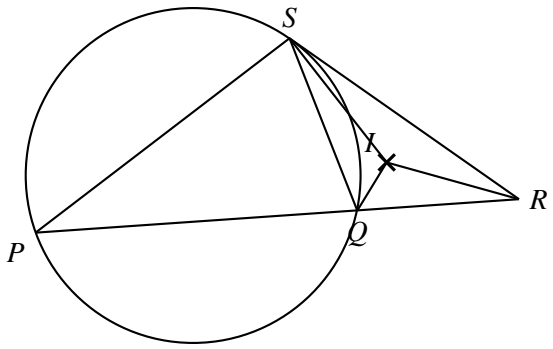
(DSE-2014 PII Q40)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. In the figure, PQS is a circle. PQ is produced to R such that RS is the tangent to the circle at S. I is the in-centre of $\triangle QRS$. If $\angle IRQ = 12^\circ$ and $\angle PSQ = 70^\circ$, then $\angle QPS =$



A. 24°.

B. 37°.

C. 43°.

D. 62°.

(DSE-2014 PII Q41)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. If the straight line $x - y = k$ and the circle $x^2 + y^2 + 2x - 4y - 1 = 0$ intersect at A and B, then the x-coordinate of the mid-point of AB is

A. $1 + k$.

B. $1 - k$.

C. $\frac{1+k}{2}$.

D. $\frac{1-k}{2}$.

(DSE-2014 PII Q42)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.